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Bridging the Knowledge Gap in Rural India

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Smart India Hackathon 2025 :

- Problem Statement ID - SIH25101
- Problem Statement Title - Remote Classrooms for Rural Community Colleges
- Theme - Smart Education
- PS Category - Software
- Team Name - EverRise
- Team ID -

Problem Statement

"Empowering Rural India with Smart, Inclusive, and Affordable Education"

- Rural students lack access to quality teachers & resources.
- Poor internet connectivity limits e-learning adoption.
- Existing platforms don't focus on local languages & rural context.
- Urban-Rural educational divide continues to grow.

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Our Solution

“Digital platform tailored for rural learners”

- **Features:**
- **Live Classes:** Attend or conduct interactive sessions
- **Study Resources:** Access notes, PDFs, videos anytime
- **Assignments & Tests:** Teachers assign, students submit
- **Chat & Attendance:** Stay engaged like a real classroom
- **Multi-language:** Hindi + English for inclusivity

Video Conferencing

Empowering students with better Education

- **Peer-to-Peer Connection –** Smooth real-time communication (100ms Live / Zego Cloud).
- **Large Class Support –** Up to 300+ students in one live lecture.
- **Real-Time Translation –** Automatic English ↔ Hindi (expandable to other languages).
- **Interactive Tools –** Raise Hand, Chat & Attendance during sessions.
- **Low-Bandwidth Friendly –** Video adjusts to weak internet for rural areas.
- **Recording Option –** Watch missed classes later.

Technology Used:

- **Frontend:** React.js, Tailwind CSS,
- **Backend:** Node.js, Express.js
- **Database:** MongoDB (Mongoose ORM)
- **Authentication:** JWT+OTP Verification
- **Real -Time:** WebRTC (Video), Socket.IO (Chat)
- **AI/ML:** NLP-based Chatbot (Dialogflow/custom ML), Recommendation Engine (Content+Collaborative filtering), Python APIs and others
- **Tools:** Figma (UI/UX), GitHub, Render, Netlify

1. RAG Chatbot Core Components :

• **LangChain** → Orchestrates the pipeline (retrieval, prompt templates, memory).

• **Hugging Face Transformers** → Provides LLMs (for text generation) and Embedding Models (for semantic search).

• **Vector Database (FAISS)** → Stores embeddings and enables fast retrieval.

2. Speech & Translation (for rural colleges)

• **Whisper (OpenAI)** → Speech-to-Text (teacher's lecture → text).

• **Hugging Face Translation Models** → Real-time translation between Hindi ↔ English (or other Indian languages).

• **Text-to-Speech (TTS) (e.g gTTS, or Hugging Face models)** → Converts translated text back into speech for students.

Impact and Future Scope:

IMPACT:

- Bridge urban–rural education divide.
- Increases access to quality content for underprivileged students.
- Empower rural youth with quality education.
- Supports Digital India & NEP 2020.

FUTURE SCOPE:

- Integration with govt. portals like DIKSHA.
- AI-driven personalized learning.
- VR/AR-based immersive learning modules.
- Expansion to skill development & job readiness programs.

Thank You

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